Motivation 2 — Lasso: convex relaxation

The standard ℓ_1 surrogate

Table of contents

> *Draft outline — content to come.*

The convex relaxation idea

Replace $\|\beta\|_0$ with $\|\beta\|_1$:

$$\min_{\beta_0,\beta} \; -\tfrac{1}{n}\,\ell(\beta_0,\beta) \;+\; \lambda\|\beta\|_1.$$

Why people use it

- Convex → solvable at scale (coordinate descent, ADMM).
- Implicit sparsity through soft-thresholding.
- Mature implementations (`glmnet`).

What it gives up

- Biased coefficients (shrinkage hits the truly active ones).
- λ-indexed, not k-indexed — the user picks a penalty, not a subset size.
- In correlated designs, tends to overshoot in selected size or pick "the wrong twin".

← Sparse GLM · Next → MIO